

Draft Rebuttal to Peer Reviews

Submission 876e

We thank all three reviewers for the careful and constructive feedback. The reviews were helpful and pointed to a common set of issues: framing, clarity of contributions, related work, quantitative evaluation, and overall presentation. In the revised manuscript, we have reorganized the paper around a clearer five-model comparison, while explicitly mapping those models into three broader representation classes. We have also rewritten the title and abstract, added a dedicated problem statement and contribution paragraph, expanded the related-work discussion, clarified the role of Model 5, strengthened the experimental protocol and results presentation, moved the calibration discussion earlier, and substantially trimmed the appendix.

For convenience, we summarize the major manuscript revisions here:

- Reframed the paper as a comparison of *five models* grouped into *three broader simplification classes*, and revised the title/abstract to match that structure.
- Reorganized the introduction to begin with the general problem, then motivate the Ising testbed, then state the paper’s question and contributions.
- Added a dedicated problem statement, explicit hypotheses, and a short contribution summary.
- Expanded the Ising background and related work, and added in-text citations clarifying which ingredients are standard and which aspects of the paper are original to this comparative study.
- Added a taxonomy table mapping each model to its representation level, strengths, and limitations; this also clarifies the role of Model 5 as a structured hybrid probability/operator surrogate.
- Strengthened the experimental section by making the calibration procedure explicit and by adding quantitative comparison metrics relative to Model 1.
- Reorganized the results to include both an overview figure and more systematic comparisons that isolate the effects of varying one parameter at a time.
- Softened or removed claims that were stronger than the evidence, especially language around “stability,” “sufficient statistics,” and “Markov blanket,” unless directly supported.
- Removed most of the philosophical appendix material from the main technical narrative and cleaned up wording, formatting, and references throughout.

Reviewer 1

Comment 1

The abstract would benefit from a brief summary of the tradeoffs among the models, for example which are more exact, less complex, or more interpretable.

Rebuttal 1

In the revision, the abstract no longer introduces the comparison only in broad terms. It now briefly summarizes the main qualitative tradeoffs among the models, including fidelity, interpretability, and computational tractability. This makes the paper’s purpose more visible from the outset and helps the reader understand why several surrogates are considered rather than only one.

Comment 2

The paper would benefit from a short background on the Ising model, especially the physical phenomenon it describes, and from a short summary of how the different modeling methods are typically used.

Rebuttal 2

We expanded the background section to better explain the Ising model as a spin system with coupling, external field, and temperature, and to clarify why it is a useful testbed for comparing simplification strategies. We also added a brief related-work discussion describing when mean-field, probability-based, and coarse-grained/operator-style approximations are typically used, so the reader can better interpret the comparison.

Comment 3

It is unclear which of the models are already known and which are new. In-text references are missing.

Rebuttal 3

We added in-text citations throughout the introduction, background, and model sections. We also now state explicitly that the novelty of the paper is primarily comparative and methodological: the main contribution is the controlled comparison of several simplification strategies within one shared experimental framework, rather than the claim that every individual model is new. Except for Model 5, which is novel and is noted as such.

Comment 4

The abstract appears to have rendered incorrectly, causing abnormal letter spacing, and the manuscript does not include an AI disclosure.

Rebuttal 4

We corrected the abstract formatting issue and performed a full proofreading pass to fix typographical inconsistencies. We also added a brief disclosure statement consistent with the course policy and clarified the role of computational tools used in drafting and revision.

Reviewer 2

Comment 1

The introduction feels slightly repetitive. It would read more cleanly if it first introduced the broader problem, then the Ising model as the testbed, then the rest of the current framing, and only after that the specific models.

Rebuttal 1

We adopted essentially this structure in the revision. The introduction now begins with the broader challenge of simplifying stochastic, high-dimensional nonlinear systems, then introduces the Ising model as a natural testbed, and only afterward presents the five models and the paper’s specific question. This removed repetition and gave the opening a clearer narrative arc.

Comment 2

The problem statement is scattered through the introduction and background. The paper would benefit from a dedicated problem statement accompanied by a hypothesis.

Rebuttal 2

The revised paper now includes a dedicated problem statement and a short list of explicit claims/hypotheses near the end of the introduction. This makes it clearer what the paper is trying to compare, what outcomes are being evaluated, and how the experiments are meant to support the paper’s conclusions.

Comment 3

Figure 1 is a good visual summary, but the results would be easier to interpret if variables were isolated more systematically, for example by holding two parameters fixed while sweeping the third.

Rebuttal 3

We retained the original summary figure as an overview, but the revised results section is now organized more systematically. In particular, we added variable-isolation plots/discussion so that the reader can see more clearly how model divergence changes as one parameter is swept while the others are held fixed.

Comment 4

It would be beneficial to add comparison metrics against Model 1, such as RMSE, as well as convergence-time figures to support the discussion that Models 3 and 4 converge more slowly.

Rebuttal 4

The revised manuscript now supplements the qualitative figure with quantitative comparisons against Model 1, including transient error and convergence-related summaries. This strengthens the claims about slower convergence and makes the differences among models more concrete than visual inspection alone.

Comment 5

Minor professionalism issues remain, including inconsistent italics and missing in-text references.

Rebuttal 5

We corrected the formatting inconsistency noted in the introduction, added the missing in-text references, and performed a broader copyediting pass to improve consistency across notation, captions, and prose.

Reviewer 3

Comment 1

The paper leaves out concrete reasons to care about the problem. Related work is limited, and the contributions are left vague.

Rebuttal 1

In the revision, the introduction more directly motivates why this comparison matters: different simplifications preserve different information and therefore support different kinds of analysis. We also added a short related-work subsection and an explicit contribution paragraph stating that the main contribution is a unified comparative framework, not a claim of a single universally best reduction.

Comment 2

No formal hypothesis is provided, even though the paper gestures toward tradeoffs among fidelity, interpretability, and cost.

Rebuttal 2

The revised manuscript now includes an explicit problem statement and a short set of hypotheses/claims describing what we expect the comparison to reveal about transient behavior, steady-state behavior, and variability across models. This makes the paper's logic easier to follow and aligns the introduction, experiments, and conclusion more directly.

Comment 3

“Stability” is mentioned in the abstract but is not really supported in the results. In addition, the title and abstract suggest three distinct linear representations, while the body compares five models, and Model 5 is especially difficult to classify.

Rebuttal 3

We revised the title, abstract, and opening sections so that the paper is now presented explicitly as a comparison of five models grouped into three broader simplification classes. We also added a taxonomy table mapping each model to its representation level, strengths, and limitations, and we explicitly describe Model 5 as a structured hybrid probability/operator surrogate rather than leaving its status implicit. Finally, we removed or softened unsupported references to “stability” and instead describe the supported results in terms of transient behavior, steady-state bias, and variance structure.

Comment 4

The results do not make novelty, tractability, and error structure sufficiently clear. The paper would be stronger if it expanded on steady-state mismatch, compute limitations, and quantitative evidence for the stated tradeoffs.

Rebuttal 4

In the revision, we clarify that the novelty is primarily methodological: the paper compares several simplification strategies under one shared protocol. We also strengthened the results section by making the discussion of mismatch more specific and by adding quantitative comparisons relative to Model 1. In addition, the revised discussion now states more directly which claims are supported by the current experiments and which remain directions for future work.

Comment 5

The first mention of the diagonal transform appears only at the end of the paper, and the future-work discussion does not justify why a genetic algorithm is the right next step.

Rebuttal 5

The calibration discussion now appears earlier, in the section on cross-model parameter translation, where we explain which transformations were fit automatically, which were adjusted manually, and why. We also revised the future-work paragraph so that it no longer introduces key methodological ideas for the first time at the end of the paper, and we softened the suggestion of a genetic algorithm from a singled-out recommendation to one possible calibration strategy among several.

Comment 6

The appendix feels disconnected from the technical paper and contributes to an overly diffuse tone.

Rebuttal 6

We substantially reduced this material and removed most of the philosophical discussion from the main submission. This keeps the focus on the technical comparison and improves the overall professionalism and coherence of the manuscript.

Comment 7

The paper feels under-polished: the conclusions do not fully match the body, wording can be clearer, and the absence of an AI disclosure raises professionalism concerns.

Rebuttal 7

We revised the conclusion so that it now reflects the actual evidence shown in the experiments rather than introducing stronger claims or new methodological details. We also completed a full wording and copyediting pass and added a disclosure statement consistent with course expectations. More broadly, the revised manuscript is now more explicit about what is claimed, what is measured, and what remains outside the scope of the current experiments. We also added an AI disclosure.

Closing remark

We again thank the reviewers for their careful reading and constructive recommendations. The revised manuscript is significantly clearer and more focused as a result. In particular, the paper now makes its framing, contribution, evaluation criteria, and limitations much more explicit, while preserving the original comparative goal of understanding how different simplifications of the same Ising dynamics preserve or distort behavior.